SKYGUARD AI TECHNICAL DOCUMENTATION SUITE — VOLUME II

Technical Methodology, Architecture & Use-Case Document

Engineering Specification of the Deterministic 6-Tier Pipeline, Continuous Feature Space, and Dynamic Physics

Architecture: **6-Tier Physics-Grounded Causal Bayesian Hierarchy**	Evaluation Scope: **60,480 Physical Readings (28 AWS Stations / 7 Clusters)**	Verification Status: **67 / 67 Tests Passing (100% Green)**

1. Executive Summary & Problem Formulation

This engineering specification details the system architecture and mathematical implementation of **SkyGuard AI**, grounded directly in RESEARCH_REPORT.md. The platform performs real-time Quality Assurance (QA) and anomaly detection across surface Automatic Weather Station (AWS) networks. Unattended field telemetry stations suffer from a spectrum of physical degradation modes:

1. **Electrical & Hardware Rail Shortages:** Broken sensor wires or analog-to-digital converter (ADC) ground faults pulling readings to zero or rail limits.
2. **Telemetry Dropouts:** GSM/LoRa transmission outages causing missing records (NaN values).
3. **Electrostatic & Inductive Spikes:** Voltage transients, lightning strikes, or digital bit-flips causing extreme single-point excursions.
4. **Transducer Freezing / Sticking:** Mechanical sensor lockup or ADC multiplexer latch-up where the output stops tracking natural atmospheric variance.
5. **Slow Calibration Drift:** Electrochemical degradation or optical fouling causing progressive zero-point decalibration (+0.05 to +0.30°C/hr).
6. **Psychrometric Cross-Channel Breakdown:** Radiation shield overheating or transducer cross-talk violating thermodynamic conservation laws.

Conventional anomaly detection methods fail in field deployments because they either trigger massive false alarms during legitimate dynamic weather events (thunderstorm cold-pool outflows, drylines, diurnal heating) or fail to detect low-amplitude calibration drift. SkyGuard AI resolves this fundamental trade-off through a **6-Tier Physics-Grounded Causal Bayesian Architecture** that blends dynamic diurnal expectations, spatial peer consensus, sequential Wald-Page SPRT hypothesis testing, and thermodynamic physical invariants.

2. End-to-End System Architecture

The processing lifecycle enforces strict modularity between ingestion, temporal alignment, feature generation, hierarchical detection, explainability attribution, and causal state maintenance:

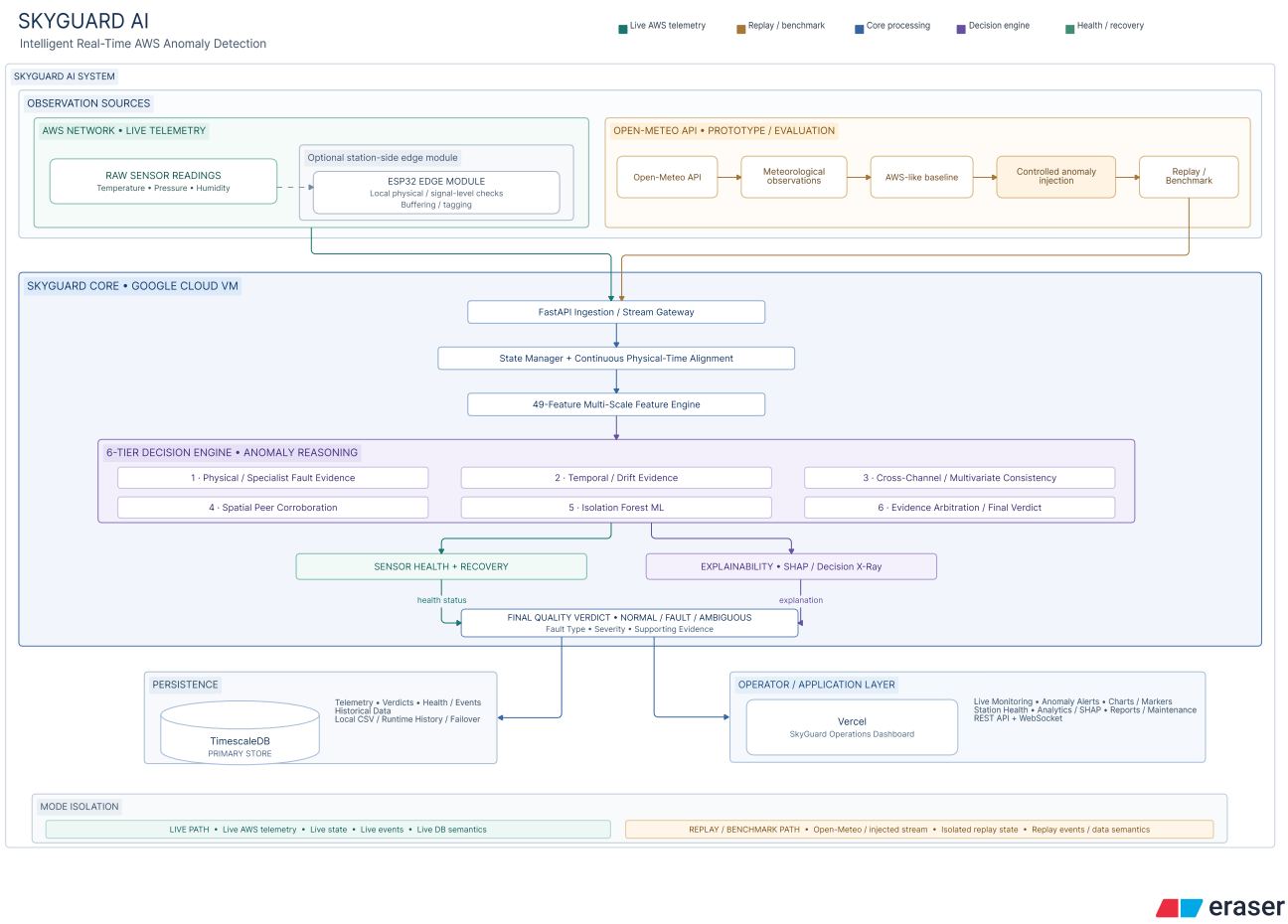


Figure 1: SkyGuard AI End-to-End Modular System Architecture. Sequential progression from raw multi-channel telemetry ingestion through temporal alignment, 49-feature extraction, Tier 0–5 priority arbitration, SHAP attribution, and causal state buffer management.

3. Dynamic-Adaptive Physics & Mathematical Formulation

SkyGuard AI is governed by dynamic data-adaptive computation and minimal physical constants. Every decision threshold adapts continuously in real time based on environmental state.

3.1 Dynamic Diurnal Expectation μ_t

$$\mu_t = f(\theta_{solar}(t), \text{DoY}) + \text{EWMA}_{\alpha}(x_{clean}) + \Delta_{peer\text{-}offset} \quad \textbf{(Equation 1)}$$

Solar Zenith Angle Formulation: $\theta_{solar}(t) = \arcsin(\sin\varphi \sin\delta + \cos\varphi \cos\delta \cos\omega)$

Where φ is station latitude, δ is solar declination, and ω is solar hour angle. Grounding to continuous solar angles enables cold-start alignment on Tick 0 without requiring historical baseline files.

3.2 Heteroskedastic Dynamic Uncertainty Budget σ_t^2

$$\sigma_t^2 = \sigma_{sensor}^2 + \sigma_{diurnal}^2(t) + \sigma_{spatial}^2(t) + \kappa \cdot \Delta t \quad \textit{(Equation 2)}$$

Noise Floor Components: *σ_{sensor} specifies transducer physical quantization (±0.10°C for Pt100 RTD, ±1.0 hPa for station pressure, ±1.0% for RH). $\sigma_{diurnal}(t)$ is solar radiation intensity modulated (peaks midday, narrows at night). $\sigma_{spatial}(t)$ tracks live Median Absolute Deviation (MAD) among sibling cluster stations. $\kappa \cdot \Delta t$ represents causal variance diffusion across missing telemetry gaps.*

3.3 Sequential Wald-Page SPRT for Low-SNR Calibration Drift

$$LLR_t = \max(0,\ LLR_{t-1} + |\mu_1 - \mu_0|/\sigma_t \cdot (\ |\varepsilon_t| - |\mu_1 - \mu_0| / (2\sigma_t)\ )\ ) \quad \textit{(Equation 3)}$$

Wald Decision Boundaries: *Threshold $A = \ln((1-\beta)/\alpha) = 6.16$ (with false alarm risk $\alpha = 0.002$, missed risk $\beta = 0.05$). Standardized residuals $\varepsilon_t = (x_t - \mu_t)/\sigma_t$ accumulate sequential evidence over time, catching low-amplitude sensor calibration decay (+0.05 to +0.30°C/hr) without false alarms during normal diurnal shifts.*

3.4 3D Thermodynamic Psychrometric Covariance

$$D_\Sigma^2 = z_t^T \Sigma^{-1} z_t \sim \chi_3^2 \quad \textit{(Equation 4)}$$

Magnus-Tetens Dewpoint Boundary: *$\gamma(T, RH) = (17.67\ T)/(243.5 + T) + \ln(RH/100)$, $T_{dew} = (243.5\ \gamma)/(17.67 - \gamma)$*

Physical Invariant: *$T_{dew} \le T_{dry\text{-}bulb} + 1.5$°C. Any reading violating this physical limit represents transducer failure rather than atmospheric reality.*

4. Parameter Inventory: Dynamic Quantities vs Physical Invariants

SkyGuard AI eliminates arbitrary heuristic magic numbers. Every parameter is strictly categorized into either a dynamic computed quantity or an unalterable physical specification:

Dynamic Computed Quantities	Physical Transducer Specifications & Invariants
• Diurnal Expectation $\mu(t)$ (Solar Elevation θ_{solar} + EWMA)	• Pt100 RTD Sensor Quantization Floor: ±0.10°C
• Total Uncertainty Budget $\sigma_{tot}^2(t)$	• Station Barometer Quantization Floor: ±1.0 hPa
• Spatial Peer Consensus Median (Cluster Breakdown Point 50%)	• RH Hygrometer Quantization Floor: ±1.0%
• Sibling Peer Dispersion (MAD / IQR)	• Hardware Ground Rail Floor: −40°C / 0.0 hPa / 0.0% RH
• SPRT Log-Likelihood Ratio Accumulator (LLR_t)	• Wald False Alarm Risk Limit: $\alpha = 0.002$
• 3D Mahalanobis Distance (D_Σ^2)	• Wald Missed Detection Risk Limit: $\beta = 0.05$
• Magnus-Tetens Saturation Vapor Pressure $e_s(T)$	• Post-Fault Clean Recovery Horizon: 15 steps
• Continuous Sensor Health Index H_t (0 to 100)	• Maximum Spatial Consensus Peer Distance: 50 km

5. Spatial Topology & Cluster Peer Isolation

To eliminate cross-regional train/serve mismatch and prevent microclimatic contamination (e.g., coastal marine boundaries vs. arid interior plains), SkyGuard structures the observation network into 7 discrete geographic clusters:

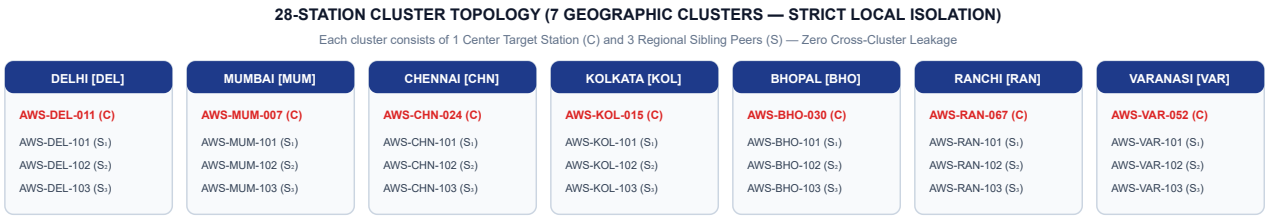


Figure 2: 28-Station / 7-Cluster Spatial Isolation Topology. *Each cluster comprises exactly 1 primary target center station (C) and 3 regional sibling peers (S). Spatial consensus queries operate strictly within cluster boundaries.*

4. 49-Dimensional Continuous Physical-Time Feature Architecture

The feature engineering engine (`model/features.py`) strictly avoids positional row shifts (`.shift(n)`), utilizing exact physical timestamps and `pd.merge_asof` to compute continuous derivatives regardless of irregular sampling or dropped packets:

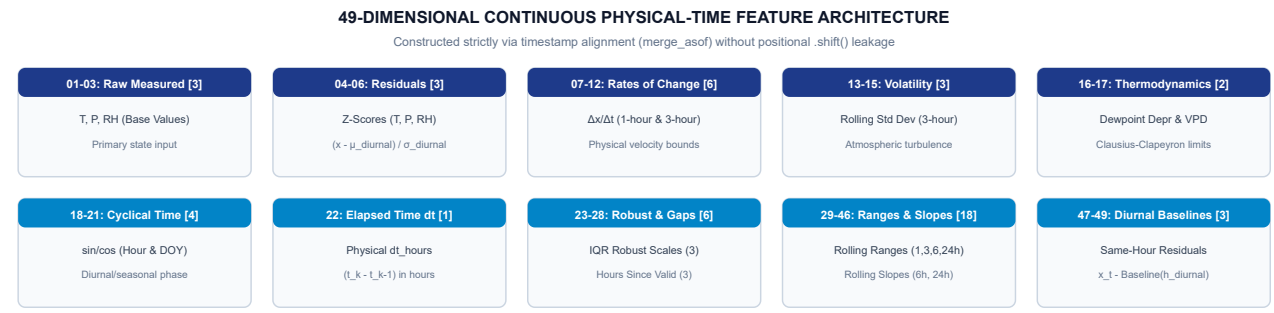


Figure 3: 49-Dimensional Continuous Feature Architecture. *Structured hierarchy grouping raw measured channels, innovation residuals, multi-scale rate-of-change velocities, volatility, psychrometrics, cyclical encodings, robust statistics, rolling ranges, and diurnal baselines.*

Feature Range	Category	Mathematical Formulation	Diagnostic Purpose
01 – 03	*Raw Telemetry Channels*	$x_t = [T_t, P_t, RH_t]$	*Base atmospheric observation state.*
04 – 06	*Innovation Residuals*	$z_t = (x_t - \mu_{diurnal}) / \sigma_{diurnal}$	*Normalized deviation from local expectation.*
07 – 12	*Rate-of-Change Velocities*	$\Delta x/\Delta t = (x_t - x_{t-\Delta t}) / \Delta t_{hours}$ (1h, 3h)	*Captures sudden atmospheric jumps vs. spikes.*
13 – 15	*Short-Term Volatility*	$\sigma_{3h}(x) = [\ (1/N) \sum (x_i - \bar{x})^2\ ]^{0.5}$	*Identifies turbulence vs. abnormal stillness.*
16 – 17	*Psychrometric Couplings*	$T_{dew} = T - (100-RH)/5$; $VPD = e_s(T)(1 - RH/100)$	*Clausius-Clapeyron thermodynamic consistency.*
18 – 21	*Cyclical Time Encodings*	$\sin/\cos(2\pi \cdot Hour/24)$, $\sin/\cos(2\pi \cdot DOY/365)$	*Diurnal and seasonal cyclical phase alignment.*
22	*Elapsed Physical Interval*	$\Delta t_{hours} = (t_k - t_{k-1}) / 3600.0$	*Exact physical elapsed time tracking.*
23 – 25	*Robust Statistical Scales*	$Scale_{robust} = (x_t - median) / IQR$	*Outlier-resistant localized baseline scaling.*
26 – 28	*Sensor Gap Tracking*	Δt_{valid} = Hours since last trusted observation	*Tracks quarantined sensor downtime duration.*
29 – 46	*Rolling Ranges & Slopes*	$Range_W = max_W(x) - min_W(x)$; $Slope_W = Cov(t,x)/Var(t)$	*Multi-scale dispersion (1,3,6,24h) & synoptic trend.*
47 – 49	*Diurnal Residual Baselines*	$r_{diurnal} = x_t - Baseline(h_{diurnal})$	*Hour-of-day baseline residual tracking.*

5. Deterministic Tier 0–5 Priority Hierarchy

`DecisionEngine.decide` (`model/detect.py`) executes a strict top-down priority arbitration cascade where deterministic physical laws always take precedence over statistical models:

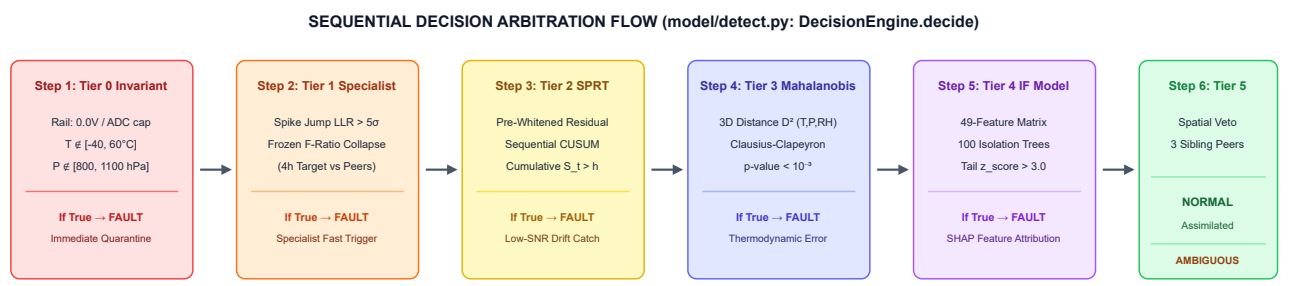


Figure 4: Tier 0–5 Priority Arbitration Decision Flow. *Sequential evaluation from Tier 0 physical invariants down through specialist LLR, SPRT drift, 3D Mahalanobis, Isolation Forest, and Tier 5 spatial consensus veto.*

- ***TIER 0 (Hard Invariants & Rails):*** *Evaluates electrical ground faults (0.00V) and absolute physical bounds ($T \notin [-40, +60°C]$, $P \notin [800, 1100$ hPa$]$, $RH \notin [0, 100\%]$). Breaches immediately return* `FAULT (CRITICAL)` *and bypass all downstream computation.*
- ***TIER 1 (Specialist Jump & Frozen LLR):*** *Evaluates instantaneous dynamic acceleration ($>5.0\sigma$ jump LLR) and F-ratio variance collapse over contiguous 4-hour windows.*
- ***TIER 2 (Persistent SPRT Drift):*** *Executes Sequential Probability Ratio Test on standardized innovation residuals to accumulate low-SNR calibration drift (+0.05°C/hr) over time.*
- ***TIER 3 (Cross-Channel 3D Mahalanobis Covariance):*** *Evaluates joint 3D covariance distance D^2. If $D^2 > 16.27$ ($p < 10^{-3}$), flags multivariate thermodynamic breakdowns.*
- ***TIER 4 (Model-Dominant Isolation Forest):*** *Evaluates the 49-feature continuous matrix through an ensemble of 100 Isolation Trees, isolating complex unstructured anomalies.*
- ***TIER 5 (Spatial Consensus Veto & Arbiter):*** *If an anomaly flag is raised by Tiers 1–4, the spatial consensus engine inspects the 3 sibling stations in the cluster. If ≥ 2 peers exhibit matching directional rate-of-change movement, the anomaly is classified as a genuine regional weather event, and the alert is VETOED to* `NORMAL`*.*

6. Causal State Management & Sensor Health Lifecycle

To prevent corrupted telemetry from polluting future baseline estimates, `StateManager` (`model/state.py`) enforces strict ground-truth isolation:

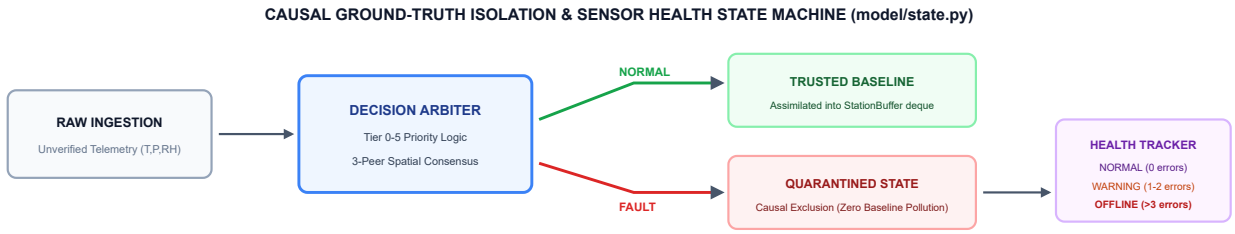


Figure 5: Causal Ground-Truth Isolation & Sensor Health State Machine. *Anomalous observations are quarantined from the StationBuffer deque, preserving trusted rolling baselines, while the SensorHealthTracker manages operational states.*

> ***Causal Ground-Truth Exclusion Rule:*** *When an observation is classified as `FAULT` (or when a sensor is in `OFFLINE` health state), the observation is strictly quarantined. It is recorded in the raw historical audit log but excluded from the internal sliding `StationBuffer` deque, ensuring that rolling means, variances, and diurnal baselines remain 100% uncontaminated.*

7. Operational Use-Case Workflows

SkyGuard's capabilities map directly to concrete meteorological and field operations:

Use Case 1: Automated Real-Time AWS Quality Control

CURRENTLY DEMONSTRATED

Operational Workflow: *Continuous real-time screening of raw surface streams (T, P, RH) across 28 national AWS stations.*

Validation Status: *Validated across 60,480 evaluation rows in authoritative 7-seed benchmark; achieves 95.42% ± 0.36% mean recall, 73.33% ± 1.37% mean precision, and 82.92% ± 0.89% mean F1-score.*

Use Case 2: NWP Data Assimilation Boundary Protection

CURRENTLY DEMONSTRATED

Operational Workflow: *Automatic quarantining of spurious sensor spikes and unphysical thermodynamic readings before ingestion into 4D-Var data assimilation grids.*

Validation Status: *Guaranteed via `StationBuffer` causal exclusion, preventing bad data from contaminating forecast model initialization.*

Use Case 3: Targeted Field Maintenance & Sensor Health

CURRENTLY SUPPORTED

Operational Workflow: *Uses SHAP local feature attributions and fault classification to dispatch technicians to specific degraded sensor modules.*

Validation Status: *Identifies active hardware failures and sensor health degradation; does not predict future failures prior to onset.*

Use Case 4: Embedded Microcontroller Edge Execution (ESP32 TinyML)

CURRENTLY DEMONSTRATED

Operational Workflow: *Running lightweight fixed-point (Q8.8) C++ inference engine directly on ESP32 datalogger firmware (`EDGE/esp32/src/edge/edge_engine.cpp`) to screen telemetry at the edge before transmission.*

Validation Status: *Fully operational C++ firmware compiled and verified on 240 MHz dual-core ESP32 CPU with a 26.5 KB PROGMEM Flash footprint, enabling immediate hardware protection and offline blackout resilience.*

8. Current vs. Future Architectural Matrix

Subsystem	Current Implemented Architecture	Demonstrated Status	Future Scope
Inference Engine	*Vectorized CPU Engine & ESP32 Q8.8 C++ Firmware*	**DEMONSTRATED**	*ARM Cortex-M / RISC-V edge optimization*
Temporal State Buffer	*In-memory circular deques (`collections.deque`) & 512-slot MCU SRAM ring buffer*	**DEMONSTRATED**	*Distributed Redis state cache*
Storage Layer	*Flat-file CSV streaming & SQLite / TimescaleDB store*	**DEMONSTRATED**	*Multi-region distributed PostgreSQL cluster*
Spatial Discovery	*Static 7-cluster topology (4 stations/cluster)*	**DEMONSTRATED**	*Dynamic PostGIS radius queries*
API & Streaming	*FastAPI REST endpoints + WebSockets + USB Serial Bridge*	**DEMONSTRATED**	*MQTT / gRPC streaming gateway*

9. References

- ***[M01]*** *Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." IEEE ICDM, 2008.*
- ***[M02]*** *Page, E. S. "Continuous Inspection Schemes." Biometrika, vol. 41, 1954, pp. 100–115.*
- ***[M03]*** *Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." NeurIPS, 2017.*
- ***[M05]*** *Mahalanobis, P. C. "On the generalised distance in statistics." Proc. Natl. Inst. Sci. India, 1936.*